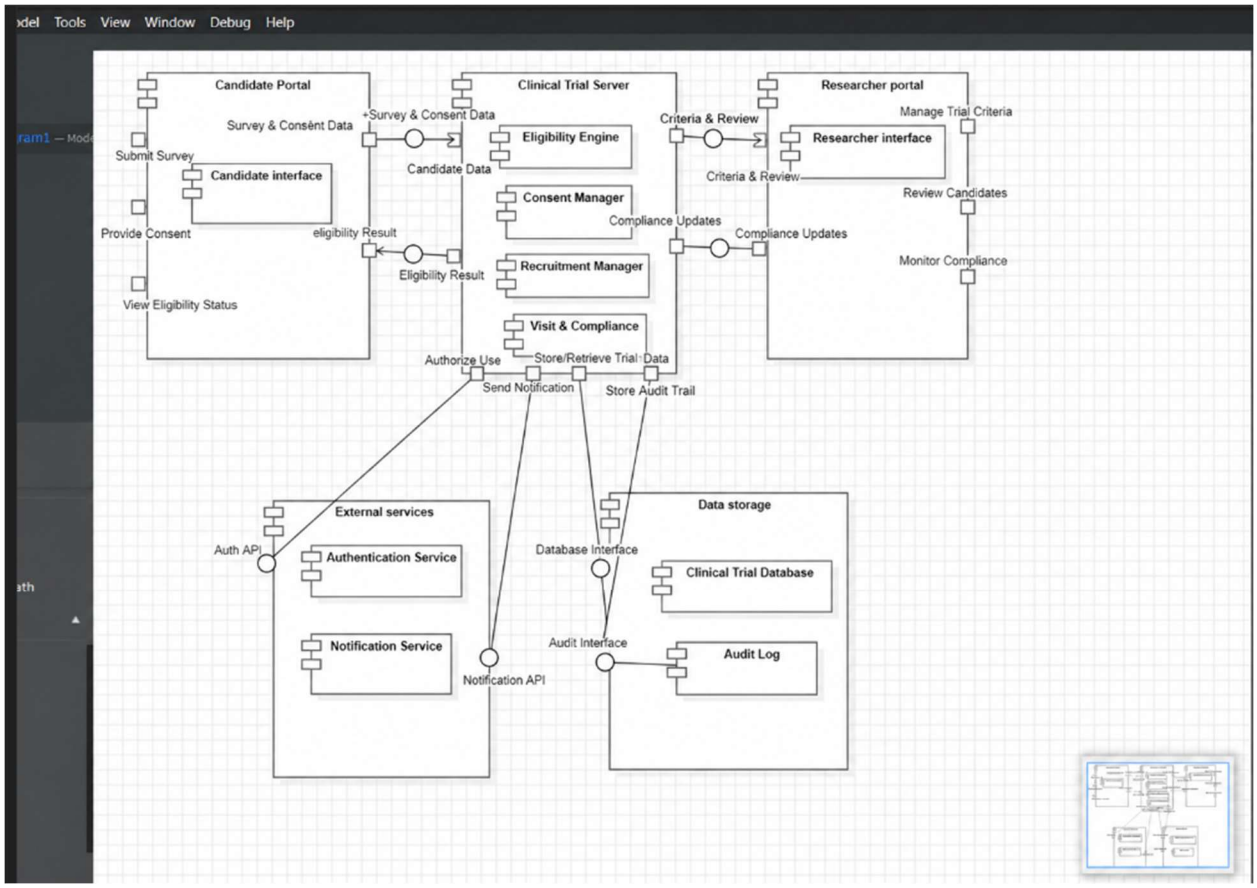
SE-LAB3

Clinical Trial Subject Recruitment Engine

Name: sham santhosh s

SRN: PES2UG24CS455



Component Diagram Explanation

The component diagram shows the main parts of our **Clinical Trial Subject Recruitment Engine** and how they communicate with each other. The system is mainly used to check whether a trial candidate is eligible for a clinical trial, manage informed consent, review candidates, and track trial participation and compliance.

1. Candidate Portal

The Candidate Portal is the part of the system used by the trial candidate. It allows the candidate to provide the required information, give informed consent, and view the eligibility result.

Candidate Interface:
It allows the candidate to submit the trial survey, provide consent, and view their eligibility status.

2. Clinical Trial Server

The Clinical Trial Server is the main processing part of the system. It handles the eligibility checking and other recruitment-related activities.

Eligibility Engine:
It checks the candidate's information against the trial's inclusion and exclusion criteria and determines the eligibility status.

Consent Manager:
It records and manages the candidate's informed consent and consent milestones.

Recruitment Manager:
It manages candidate recruitment information and supports the process of reviewing and tracking candidates.

Visit & Compliance:
It records trial visits and monitors whether the candidate follows the required visit and compliance schedule.

3. Researcher Portal

The Researcher Portal is used by the Clinical Researcher to manage the trial and review candidate information.

Researcher Interface:
It allows the researcher to manage trial criteria, review candidate eligibility, and monitor compliance information.

4. External Services

External Services provide common services required by the system.

Authentication Service:
It verifies users and controls access to the recruitment system.

Notification Service:
It sends notifications related to recruitment, consent milestones, visits, and other important updates.

5. Data Storage

Data Storage keeps the important information generated by the system.

Clinical Trial Database:
It stores trial criteria, candidate information, eligibility results, consent information, visit details, and recruitment records.

Audit Log:
It records important system activities and changes for traceability and supports the **21 CFR Part 11** electronic-record and audit-trail requirement.